



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to JEYAM FUELS

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

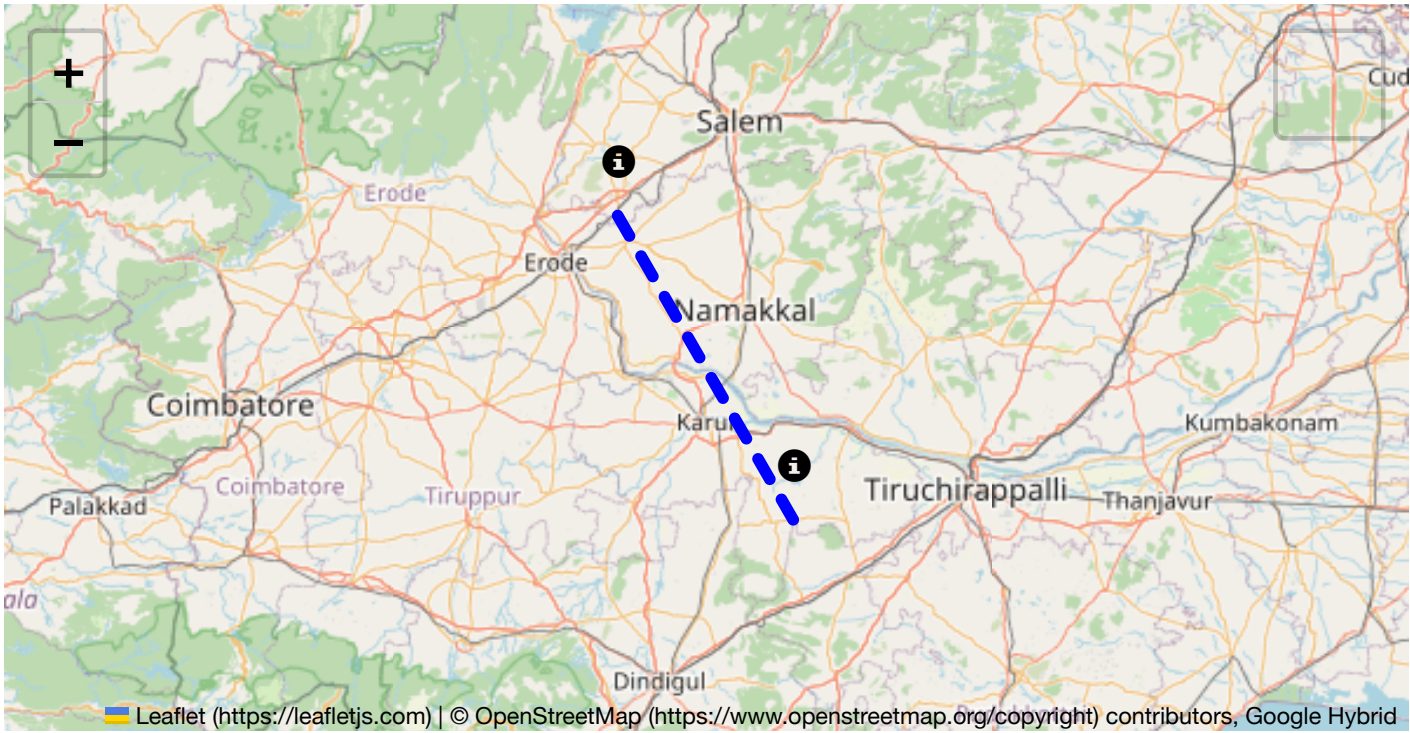
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 110.41 km
Estimated Duration: 2.3 hours
Adjusted Duration (Heavy Vehicle): 2.8 hours

Start: (11.4381, 77.8734)
End: (10.7344537, 78.2825014)



Welcome to the Journey Risk Management Study

Route Safety Analysis from Sangagiri to Mathagiri, Tamil Nadu

- Overview of the Route Map:**
 - The route from Sangagiri to Mathagiri covers approximately 110.41 kilometers, primarily traversing through state highways and parts of the National Highway (NH44). The journey generally takes about 2.27 hours for heavy vehicles carrying hazardous materials. This route passes through various towns and rural areas with varying road conditions and traffic densities.
- Typical Weather Conditions and Potential Weather-Related Hazards:**
 - Tamil Nadu experiences tropical climate conditions with dry summers (March to June) and heavy monsoon rains (July to September, and a shorter northeast monsoon between October and December). Potential weather-related hazards include heavy rains leading to waterlogging, reduced visibility, and occasionally flooding, particularly during the monsoon months.
- Traffic Patterns and Congestion-Prone Areas:**
 - Peak traffic hours in Tamil Nadu are typically between 8:00-10:00 AM and 5:00-7:00 PM. Congestion is common in urban centers and near popular intersections. The route from Sangagiri to Mathagiri may experience peak traffic in towns such as Salem and Dharmapuri. NH44 can also have heavy truck traffic and occasional slow-moving vehicles.
- Assessment of Road Quality and Infrastructure:**
 - State highways and NH44 are generally well-maintained, but conditions can vary drastically. Potholes or uneven surfaces may be present, especially after heavy rains. Certain rural stretches might lack robust signage or lighting.
- Suggestions for Alternative Routes for Emergencies:**
 - In case of emergencies or road blockages, alternative routes could involve detours via state highways that connect back to NH44. Access local maps for smaller, connecting roads around

Salem for detours.

6. Local Regulations Affecting Hazardous Material Transport:

- Adhere to regulations as per the Tamil Nadu Motor Vehicle Rules and the Central Motor Vehicles Rules, which include restrictions on times of transport, vehicle markings, and documentation required for transporting hazardous materials.

7. Overview of Historical Incidents:

- There have been occasional incidents involving heavy vehicles on NH44, typically due to brake failures, speeding or weather conditions causing accidents. Areas around Salem have recorded traffic incidents due to its busy traffic junctures.

8. Environmental Considerations and Sensitive Areas:

- The route passes through agricultural and populated areas, where spillage of hazardous materials can impact the environment and local communities. Care should be taken to prevent leakage or spillage.

9. Communication Coverage:

- While most of the route should have good mobile coverage, there might be sporadic dead zones in rural areas. It is advisable to ensure seamless communication by checking with local carriers for coverage maps.

10. Estimated Emergency Response Times:

- Urban areas like Salem may have quicker emergency response times (approx. 15-30 minutes), while rural areas can experience delays (30 minutes to over an hour), especially at night or during adverse weather conditions.

11. Summary of Risk Assessment:

- Overall, this route is moderately safe, though it does pose risks associated with variable weather, traffic congestion, and road conditions. Preparedness for adverse weather, ensuring adherence to transport regulations, and monitoring traffic and road conditions can help mitigate risks effectively. Communication plans and knowledge of emergency procedures are crucial for ensuring safety during transportation of hazardous materials.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.38395, 77.89480	15 KM/Hr
1	Roundabout	High	10.92958, 78.05452	15 KM/Hr
2	U-Turn	High	10.9110477, 78.09713769999999	10 KM/Hr
3	Turn	High	11.43968, 77.87345	15 KM/Hr
4	Turn	High	11.44019, 77.87547	15 KM/Hr
5	Turn	Medium	11.39685, 77.88861	30 KM/Hr
6	Turn	High	11.37862, 77.89488	15 KM/Hr
7	Turn	High	11.37850, 77.89453	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
8	Turn	Medium	11.16950, 78.01544	30 KM/Hr
9	Turn	Medium	11.16764, 78.01565	30 KM/Hr
10	Turn	High	11.14545, 78.02208	15 KM/Hr
11	Turn	High	11.14480, 78.02291	15 KM/Hr
12	Turn	Medium	11.10102, 78.00572	30 KM/Hr
13	Turn	Medium	11.05261, 78.05138	30 KM/Hr
14	Turn	Medium	10.90767, 78.09893	30 KM/Hr
15	Turn	High	10.90793, 78.09916	15 KM/Hr
16	Blind Spot	Blind Spot	10.91105, 78.09714	10 KM/Hr
17	Turn	High	10.84613, 78.12000	15 KM/Hr
18	Turn	Medium	10.84321, 78.16436	30 KM/Hr
19	Turn	High	10.84057, 78.16518	15 KM/Hr
20	Turn	High	10.83953, 78.18108	15 KM/Hr
21	Turn	Medium	10.83523, 78.18173	30 KM/Hr
22	Turn	Medium	10.82892, 78.20166	30 KM/Hr
23	Turn	Medium	10.82865, 78.20188	30 KM/Hr
24	Turn	Medium	10.80081, 78.20092	30 KM/Hr
25	Turn	Medium	10.79274, 78.20214	30 KM/Hr
26	Turn	Medium	10.79217, 78.20276	30 KM/Hr
27	Turn	Medium	10.79013, 78.20307	30 KM/Hr
28	Turn	Medium	10.78767, 78.20438	30 KM/Hr
29	Turn	Medium	10.78380, 78.20278	30 KM/Hr
30	Turn	Medium	10.78037, 78.20649	30 KM/Hr
31	Turn	High	10.77911, 78.21461	15 KM/Hr
32	Turn	Medium	10.77330, 78.21431	30 KM/Hr
33	Turn	High	10.77239, 78.21467	15 KM/Hr
34	Turn	High	10.77215, 78.21560	15 KM/Hr
35	Turn	High	10.76401, 78.21632	15 KM/Hr
36	Turn	Medium	10.75296, 78.22112	30 KM/Hr
37	Turn	Medium	10.75178, 78.22246	30 KM/Hr
38	Turn	Medium	10.74981, 78.22308	30 KM/Hr
39	Turn	Medium	10.74551, 78.22709	30 KM/Hr
40	Turn	Medium	10.73084, 78.23146	30 KM/Hr
41	Turn	High	10.72910, 78.23136	15 KM/Hr
42	Turn	Medium	10.72872, 78.23407	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Tiruchengode, Goverment Hospital	11.3903328, 77.8920627	30 km/h	Medium
1	hospital	SPM Medical Centre,Tiruchengode	11.3881331, 77.8931963	30 km/h	Medium
5	hospital	T.C.A Hospital Tiruchengode	11.3791885, 77.8965774	30 km/h	Medium
6	clinic	Kongu Nursing Home	11.3783065, 77.8961134	30 km/h	Medium
7	hospital	Soorya Multispecialty Hospital	11.3786429, 77.8931912	30 km/h	Medium
8	hospital	Tiruchengode Government Hospital	11.37645, 77.89426	30 km/h	Medium
9	hospital	Krishna Hospital, Namakkal	11.3754811, 77.8931817	30 km/h	Medium
10	hospital	Tirukumaran Hospitals	11.3782118, 77.8914933	30 km/h	Medium
13	police	கரட்டுப்பாளையம் காவல் நிலையம்	11.357586, 77.8922539	30 km/h	Medium
14	hospital	Goverment Hospital, Nallur	11.2679698, 77.946863	30 km/h	Medium
16	hospital	Goverment Hospital	11.156647, 78.0213586	30 km/h	Medium
19	hospital	Mylampatty Government Hospital	10.7320815, 78.2459642	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
2	school	அரசு ஆண்கள் மேல்நிலைப் பள்ளி	11.3850439, 77.8948279	30 km/h	Medium
3	school	அரசு பெண்கள் மேல்நிலைப் பள்ளி	11.3844247, 77.8949053	30 km/h	Medium
4	marketplace	திருச்செங்கோடு தினசரி காய்கறி சந்தை	11.3833608, 77.8970145	30 km/h	Medium
11	school	KSR Educational institution	11.3772013, 77.8908807	30 km/h	Medium
12	school	MDV School	11.3719843, 77.8915244	30 km/h	Medium
15	school	Sivabakkiam Muthusamy Matriculation Higher Secondary School	11.1788177, 78.0061696	30 km/h	Medium
17	school	Goverment Higher Secondary School	11.1511724, 78.0200723	30 km/h	Medium
18	college	valluvar catering	10.9553473, 78.0606583	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.38395, 77.89480



Risk Type: Roundabout
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 10.92958, 78.05452



Risk Type: U-Turn
Risk Level: High
Speed Limit: 10 KM/Hr
Coordinates: 10.9110477, 78.09713769999999



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44019, 77.87547



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.39685, 77.88861



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.37862, 77.89488



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.37850, 77.89453



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.16950, 78.01544



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.16764, 78.01565



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.14545, 78.02208



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.14480, 78.02291



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.10102, 78.00572



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.05261, 78.05138



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.90767, 78.09893



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 10.90793, 78.09916



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 10.91105, 78.09714



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 10.84613, 78.12000



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.84321, 78.16436



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 10.84057, 78.16518



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 10.83953, 78.18108



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.83523, 78.18173



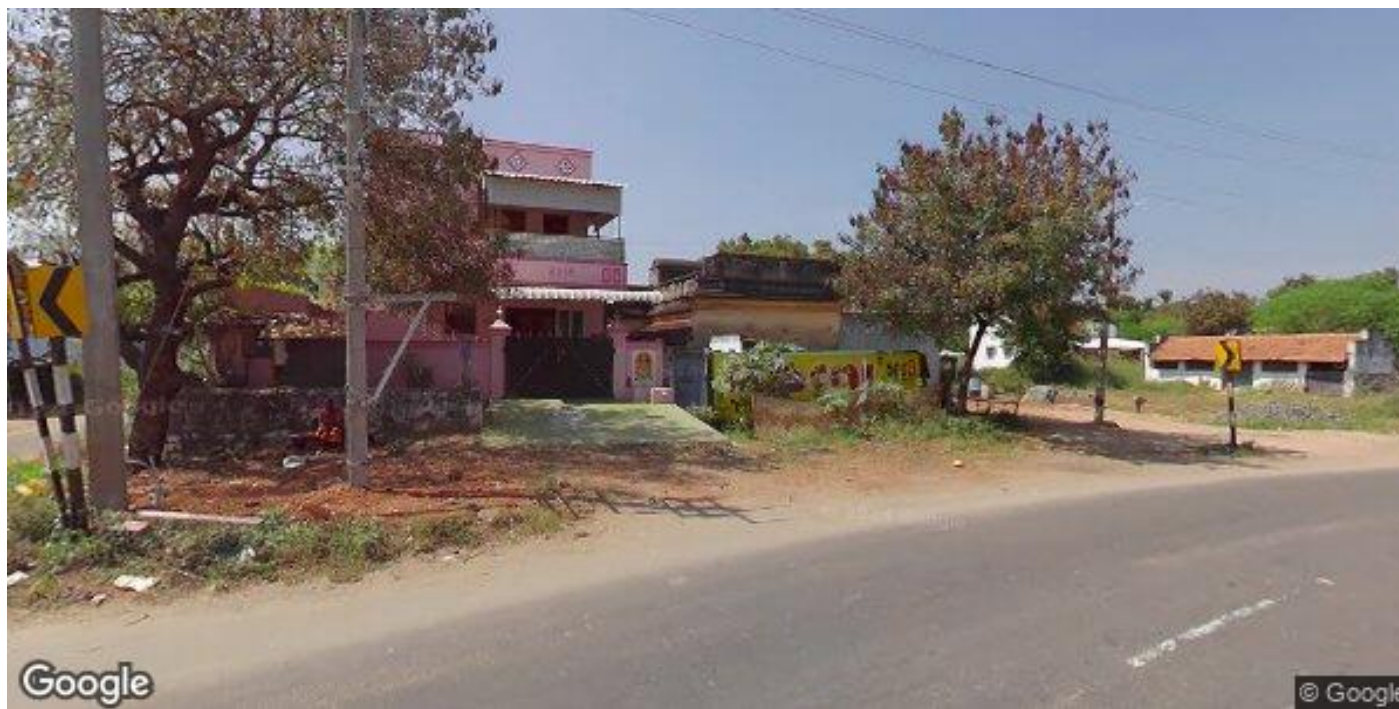
Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.82892, 78.20166



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.82865, 78.20188



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.80081, 78.20092



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.79274, 78.20214



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.79217, 78.20276



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.79013, 78.20307



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.78767, 78.20438



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.78380, 78.20278



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.78037, 78.20649



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 10.77911, 78.21461



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.77330, 78.21431



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 10.77239, 78.21467



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 10.77215, 78.21560



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 10.76401, 78.21632



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.75296, 78.22112



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.75178, 78.22246



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.74981, 78.22308



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.74551, 78.22709



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.73084, 78.23146



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 10.72910, 78.23136



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.72872, 78.23407

